

ARCHITECTURE CONCEPT

The Invisible

Translating Unstructured Human Dialogue into Structured Execution Trees

(Intent AST)

Compiler

CORE CHALLENGE

From Dialogue to Determinism

Human speech is messy, implicit, and context-dependent. Machine execution requires rigid structure, clarity, and predictable state transitions.

Bridging Conversation & Logic

INPUT PHASE

Conversational Input

Natural conversation contains ambiguity, missing metadata, implied assumptions, and informal language. Humans rely on context to fill in the gaps.

"Hey, can you clean up the kitchen before dinner?"

COMPILATION PHASE

The Invisible Compiler

Silently running in the background, the Invisible Compiler parses dialogue into an **IntentAST**—a structured plan ready for automated execution.

Parse → Extract Nodes → Build Graph → Validate → Execute

TREE STRUCTURE

Anatomy of an IntentAST

An Abstract Syntax Tree for Intent (IntentAST) breaks any conversation down into four foundational, deterministic nodes.

The Four Core Nodes



NODE 1

Actor

Who is designated to initiate or execute the action? (User, AI Agent, TA, Apprentice)



NODE 2

Context

What are the environment, constraints, safety rules, and operational boundaries?



NODE 3

Action

What atomic, step-by-step operations must be carried out sequentially?



NODE 4

Outcome

What is the verifiable end state or expected post-condition?

Example 1: Kitchen Safety

DIALOGUE

”Head Chef to Apprentice: Fry the potatoes safely before the dinner rush.”

- **Actor:** Apprentice Chef
- **Context:** Commercial fryer, 350°F oil, heat gloves required, prep time window
- **Action:** Lower batch into hot oil; set 4-minute timer; drain basket
- **Outcome:** Golden crisp fries served safely with zero spill incidents



Example 2: Classroom Planning

DIALOGUE

”Teacher to Assistant: Prep 25 lab stations before 9 AM chemistry class.”

→ **Actor:** Teaching Assistant (TA)

→ **Context:** Science Lab Room 204, 15 minutes before 1st bell, 25 students

→ **Action:** Distribute safety goggles, place 25 beaker sets on desks

→ **Outcome:** All stations equipped; class starts immediately without delay

Lessons at Our School	
In line with our framework for teaching and learning, this table outlines the features of lessons that should be common to all teachers in all subjects, except where there are special requirements for learning in that department.	
Interacting with Student Data	All teachers should have on-hand or up-to-date data plan for every class they teach. This can be a seating plan or any helpful hard-copy format. Data can be coded but should reference VSD data or Starting Profile, SDNYAL status, housing, Aps, AP status, IB/AP status and latest attainment grade.
Routines for All Lessons	
Starting Lessons: Entry Routines	- Teacher welcomes class at the door; they go straight in without talking, sit down and get their books and equipment out ready to learn, engaging with any written instructions provided. - Teacher uses signal for attention and addresses class with full attention, setting expectations for introductory activities. - Once students are working, register is taken. Registers should be taken close to the very beginning with a full 100% where this defaults as an open register. - If students arrive before the teacher, they wait until the teacher arrives and the teacher acknowledges their presence.
Behaviour for Learning	Behavioural consequences must be logged visibly on the board and the teacher publicly reminded to reinforce expectations for all students. Consequences should be issued immediately right from the start. No disruption should be tolerated. On Call contacted as required.
Showing Excellence and Positive Affirmation	At least five minutes in every lesson should be devoted to showcasing examples of excellent work or attitudes to learning, highlighting the reasons. Achievement points should be issued publicly at this time.
Ending Lessons: Exit Routines	Students stand behind their chairs with all equipment packed away. Teacher dismisses them from the door, table by table, calmly into the corridor on the plan.
Common Pedagogical Elements	
Specifying knowledge elements and Retrieval Practice	It is essential that teachers make very explicit all aspects of the content explored in the lesson that are required knowledge. Use of knowledge organisers will help this. Regular questioning and other forms of retrieval practice should be routine.
Modeling and Practice	Where new ideas or new skills are being introduced, teachers should always model the work expected from students. This could be through multiple worked examples, student exemplars or demonstrations. Students must have time to practice skills repeatedly.
Structured, targeted questioning	Questioning should include all students with answers selected by the teacher in a deliberate, planned manner. Questioning should be probing and targeted to specific students where appropriate. Students should not have the option to opt out or to dominate.
Responding to Feedback	Feedback will take many forms – verbal comments, written comments, peer and self-assessment. There should be evidence that feedback leads to students' work improving or changing. Agreed departmental workflow procedures should be followed. Students and teachers should all be clear about where and when feedback will be given and which work should be redrafted, improved or corrected.
Features of Good Speech	Students should be required to adhere to the guidance on good speech. This applies to general discussion as well as to structured speech events.
Homework	Homework should be set each week. All homework must be recorded on the online platform, issued during the lesson and collected the same day.

IntentAST Node Mapping

Domain	Actor	Context	Action	Outcome
Kitchen Prep	Apprentice Chef	Station #3, 350°F Oil, Wear Gloves	Fry Potato Batch (4 mins)	Crispy Fries, Clean Station
Classroom	Teaching Assistant	Room 204, Pre-Class Window	Set Up 25 Chemistry Kits	Ready for 9 AM Lesson
Software Support	Automated Bot	Active User Session, Auth Token	Reset Password & Send Link	Account Recovered Securely

Why Compile Intent?

100%

Deterministic Plans

SYSTEM RELIABILITY

Eliminating Hallucinations

By compiling casual conversation into explicit IntentAST nodes, automated systems eliminate guesswork and ambiguity. Agents no longer hallucinate parameters—they execute validated, structured steps according to strict contextual rules.

KEY TAKEAWAY

Conversation is Code

The Invisible Compiler turns everyday dialogue into reliable, structured workflows.

Actor • Context • Action • Outcome

Image Sources



<https://burncenters.com/wp-content/uploads/2022/01/NBAW22-Commercial-BLOG.jpg>

Source: burncenters.com



<https://teacherhead.com/wp-content/uploads/2017/10/screen-shot-2017-10-08-at-11-33-17.png>

Source: teacherhead.com